Supporting information for: Koopmans spectral functionals in periodic-boundary conditions

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KI matrix elements

For clarity and completeness, we provide here additional details on the KI matrix elements and on the derivation of Eqs. (20) and (22) in the main text. Let us start with the definition

of the KI matrix element on two Wannier functions $|\mathbf{R}i\rangle$ and $|\mathbf{0}j\rangle$:

$$\begin{aligned}\Delta H_{ij}^{\mathrm{KI}(2)}(\mathbf{R}) &= \langle \mathbf{R}i|\hat{\mathcal{V}}_{\mathbf{0}j}^{\mathrm{KI}(2)}|\mathbf{0}j\rangle \\ &= \int d\mathbf{r}\omega_{\mathbf{R}i}^{*}(\mathbf{r})\mathcal{V}_{\mathbf{0}j}^{\mathrm{KI}(2)}(\mathbf{r})\omega_{\mathbf{0}j}(\mathbf{r}) \end{aligned} \tag{1}$$

From a computational point of view it is convenient to first evaluate the product of the two Wannier functions in real space and then perform the integral with the KI potential in reciprocal space. Using the definition of Wannier functions in Eq. (8) of the main text, the product of Wannier functions is given by

$$\begin{aligned}\omega_{\mathbf{R}i}^{*}(\mathbf{r})\omega_{\mathbf{0}j}(\mathbf{r}) =& \frac{1}{N_{\mathbf{k}}N_{\mathbf{k}'}}\sum_{\mathbf{k}\mathbf{k}'} e^{i\mathbf{k}\cdot\mathbf{R}}\tilde{\psi}_{\mathbf{k}i}^{*}(\mathbf{r})\tilde{\psi}_{\mathbf{k}'j}(\mathbf{r}) \\ =& \frac{1}{N_{\mathbf{k}}N_{\mathbf{q}}}\sum_{\mathbf{k}\mathbf{q}} e^{i\mathbf{k}\cdot\mathbf{R}}e^{i\mathbf{q}\cdot\mathbf{r}}\tilde{u}_{\mathbf{k}i}^{*}(\mathbf{r})\tilde{u}_{\mathbf{k}+\mathbf{q}j}(\mathbf{r}) \\ =& \frac{1}{N_{\mathbf{k}}}\sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{R}}\frac{1}{N_{\mathbf{q}}}\sum_{\mathbf{q}} e^{i\mathbf{q}\cdot\mathbf{r}}\rho_{\mathbf{k},\mathbf{k}+\mathbf{q}}^{ij}(\mathbf{r}), \end{aligned} \tag{2}$$

while the KI potential is given by a purely scalar term $-\frac{1}{2}\Delta_{\mathbf{0}j}^{\mathrm{KI}(2)}$ plus an $\mathbf{r}$-dependent contribution (only for empty states)[S1] that can be decomposed into a sum of monochromatic terms:

$$\mathcal{V}_{\mathbf{0}j}^{\mathrm{KI}(2)}(\mathbf{r}) = -\frac{1}{2}\Delta_{\mathbf{0}j}^{\mathrm{KI}(2)} + (1-f_j)\sum_{\mathbf{q}} e^{i\mathbf{q}\cdot\mathbf{r}}V_{\mathrm{pert},\mathbf{q}}^{\mathbf{0}j}(\mathbf{r}) \tag{3}$$

Combining Eq. 2 and 3 and using the orthogonality property of the Wannier functions $\langle \mathbf{R}i|\mathbf{0}j\rangle = \delta_{\mathbf{R}\mathbf{0}}\delta_{ij}$, we get

$$\Delta H_{ij}^{\mathrm{KI}(2)}(\mathbf{R}) = -\frac{1}{2}\Delta_{\mathbf{0}j}^{\mathrm{KI}(2)}\delta_{\mathbf{R}\mathbf{0}}\delta_{ij} + \Delta H_{ij,\mathbf{r}}^{\mathrm{KI}(2)}(\mathbf{R}) \tag{4}$$

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where $\Delta_{\mathbf{0}j}^{\mathrm{KI}(2)}$ is the scalar term and it is simply given by the self Hartree-exchange-correlation of the Wannier orbital density

$$\begin{aligned}\Delta_{\mathbf{0}j}^{\mathrm{KI}(2)} &= \langle\rho_{\mathbf{0}j}|\left[f_{\mathrm{Hxc}}\right]|\rho_{\mathbf{0}j}\rangle = \langle\rho_{\mathbf{0}n}|V_{\mathrm{pert}}^{\mathbf{0}n}\rangle \\ &= \frac{1}{N_{\mathbf{q}}}\sum_{\mathbf{q}}\langle\rho_{\mathbf{q}}^{\mathbf{0}j}|V_{\mathrm{pert},\mathbf{q}}^{\mathbf{0}j}\rangle. \end{aligned} \tag{5}$$

and $\Delta H_{ij,\mathrm{r}}^{\mathrm{KI}(2)}(\mathbf{R})$ is the contribution from the $\mathbf{r}$-dependent part of the potential:

$$\begin{aligned}\Delta H_{ij,\mathbf{r}}^{\mathrm{KI}(2)}(\mathbf{R}) &= (1-f_j)\frac{1}{N_{\mathbf{k}}}\sum_{\mathbf{k}}e^{i\mathbf{k}\cdot\mathbf{R}}\frac{1}{N_{\mathbf{q}}^2}\sum_{\mathbf{q}\mathbf{q}'}\int d\mathbf{r}e^{i(\mathbf{q}+\mathbf{q}')\cdot\mathbf{r}}V_{\mathrm{pert},\mathbf{q}}^{\mathbf{0}j}(\mathbf{r})\rho_{\mathbf{k},\mathbf{k}+\mathbf{q}'}^{ij}(\mathbf{r}) \\ &= (1-f_j)\frac{1}{N_{\mathbf{k}}}\sum_{\mathbf{k}}e^{i\mathbf{k}\cdot\mathbf{R}}\frac{1}{N_{\mathbf{q}}^2}\sum_{\mathbf{q}\mathbf{q}'}\sum_{\mathbf{g}\mathbf{g}'}\int d\mathbf{r}e^{i(\mathbf{q}+\mathbf{q}')\cdot\mathbf{r}}e^{i(\mathbf{g}+\mathbf{g}')\cdot\mathbf{r}}V_{\mathrm{pert},\mathbf{q}}^{\mathbf{0}j}(\mathbf{g})\rho_{\mathbf{k},\mathbf{k}+\mathbf{q}'}^{ij}(\mathbf{g}') \\ &= (1-f_j)\frac{1}{N_{\mathbf{k}}}\sum_{\mathbf{k}}e^{i\mathbf{k}\cdot\mathbf{R}}\left\{\frac{1}{N_{\mathbf{q}}}\sum_{\mathbf{q}\mathbf{g}}V_{\mathrm{pert},\mathbf{q}}^{\mathbf{0}j}(\mathbf{g})\rho_{\mathbf{k},\mathbf{k}-\mathbf{q}}^{ij}(-\mathbf{g})\right\} \qquad [\mathbf{q}\to-\mathbf{q},\mathbf{g}\to-\mathbf{g}] \\ &= (1-f_j)\frac{1}{N_{\mathbf{k}}}\sum_{\mathbf{k}}e^{i\mathbf{k}\cdot\mathbf{R}}\left\{\frac{1}{N_{\mathbf{q}}}\sum_{\mathbf{q}\mathbf{g}}V_{\mathrm{pert},-\mathbf{q}}^{\mathbf{0}j}(-\mathbf{g})\rho_{\mathbf{k},\mathbf{k}+\mathbf{q}}^{ij}(\mathbf{g})\right\} \\ &= \frac{1}{N_{\mathbf{k}}}\sum_{\mathbf{k}}e^{i\mathbf{k}\cdot\mathbf{R}}\left\{(1-f_j)\frac{1}{N_{\mathbf{q}}}\sum_{\mathbf{q}\mathbf{g}}[V_{\mathrm{pert},\mathbf{q}}^{\mathbf{0}j}(\mathbf{g})]^*\rho_{\mathbf{k},\mathbf{k}+\mathbf{q}}^{ij}(\mathbf{g})\right\}. \end{aligned} \tag{6}$$

The term inside the braces can be readily identified with the KI Hamiltonian at the given $\mathbf{k}$-point and is the final results for the KI matrix elements reported in Sec. 3.1.2. From a computational point of view, the product of the (periodic part of the) Wannier functions $\rho_{\mathbf{k},\mathbf{k}+\mathbf{q}}^{ij}$ is performed in real space and then Fourier-transformed to the reciprocal space where the scalar product with the KI potential $V_{\mathrm{pert},\mathbf{q}}^{\mathbf{0}j}$ is performed. Each independent $\mathbf{q}$-contribution is then summed up to give the desired matrix element.

Full versus 2nd order approximation

In this section we provide additional tests for the second order approximation adopted in this work for the KI energy and potential corrections. As a reference we take a full KI

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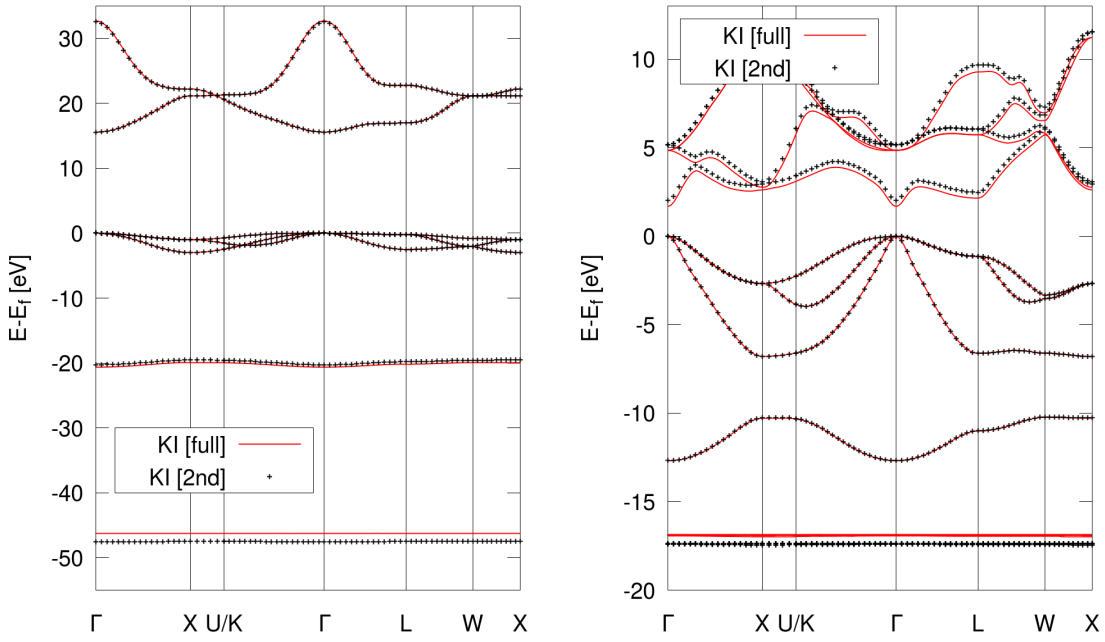


Figure S1: KI Band structure of LiF (left panel) and GaAs (right panel) calculated. Solid red lines are the results from a full KI calculation as described in Ref. S2, black cross are the results of an equivalent calculation within the present implementation and approximation.

calculation performed as detailed in Ref. S3 and Ref. S2 where i) the screening coefficients are computed with a finite difference approach removing/adding one electron to the system (and not just a tiny fraction as in the 2nd order approach presented here), and ii) the KI energy and potential corrections are not approximated to 2nd order. We instead resort to the approximation of using MLWFs as variational orbitals also for the SC calculations presented here.

In Fig. S1 we show a comparison between full KI band structure calculations done in a super-cell plus unfolding implementation[S2] and the present approach for LiF and GaAs. We used the same computational set-up used in Ref. S2, i.e. a 4×4×4 mesh for the sampling of the Brilloiun zone to match the dimension of the supercell used in Ref. S2, the same lattice constants, base functional (PBE) and kinetic-energy cutoff for the expansion of the wave functions. For GaAs the smooth interpolation method presented in Ref. S2 is also used to improve the quality of the band interpolation and to have a fair comparison with

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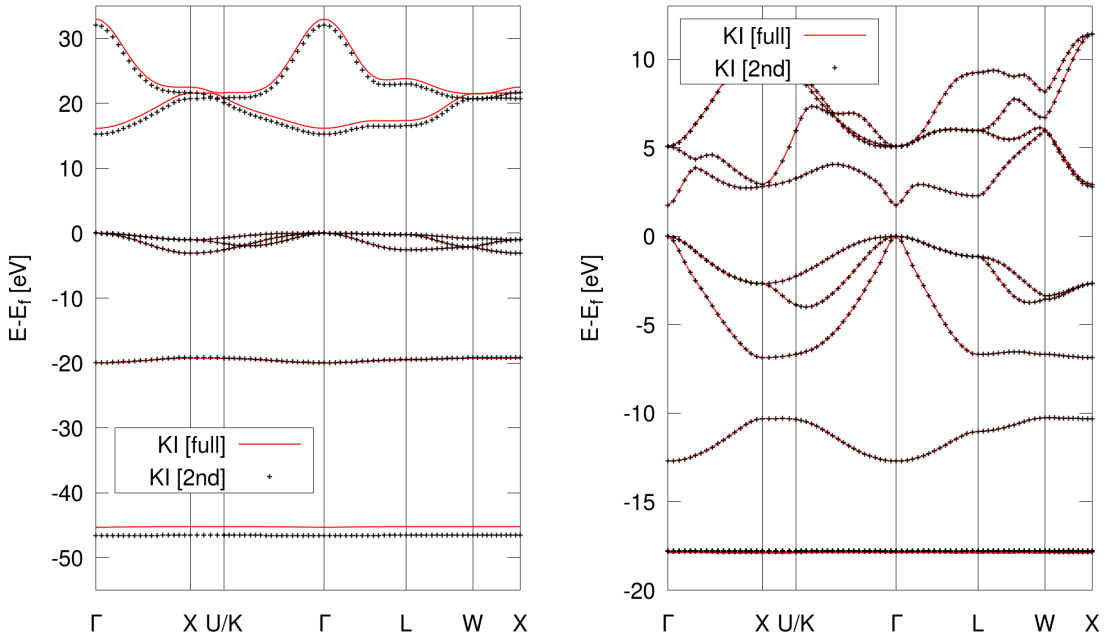


Figure S2: As in Fig. S1 but using LDA as the base functional.

the reference results from Ref. S2. For LiF we have an overall very good agreement between the two calculations; the valence and conduction bands are in perfect agreement with the reference full KI calculation, both in terms of energy position and shape. For the F 2s and Li 1s bands, lying at about -20 eV and -47 eV with respect to the top of the valence band, we observe a difference of 0.3 eV (1.5 %) and 1.3 eV (2.8 %), respectively. For GaAs the differences are more marked with a 0.3 eV difference in the band gap and 0.5 eV in the position of the Ga d bands. With respect to experiment, this leads to a slightly worse agreement for the band gap and a slightly better one for the Ga d energy.

As an additional check we performed the same comparison as in Fig. S1, but now using the LDA as the base functional (the same used in the main text). The comparison is shown in Fig. S2. For GaAs the agreement between the full and the 2nd order approximation is almost perfect, as was already observed for the density of state in Fig. 2 of the main text. At the same time we observe a slightly worse agreement for LiF, in particular the band gap is underestimated by 0.8 eV (5%) compared to the full KI results. A summary of the energy

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Table S1: Relevant energy values for LiF and GaAs from full and 2nd order KI calculations. The zero of the energy is set to the VBM. KIPZ values for LiF[S2] are also reported for comparison.

		LDA			PBE		
		E_g	$\langle\varepsilon_{F_{2s}}\rangle$	$\langle\varepsilon_{Li_{1s}}\rangle$	E_g	$\langle\varepsilon_{F_{2s}}\rangle$	$\langle\varepsilon_{Li_{1s}}\rangle$
LiF	KI Full	16.16	-19.6	-45.2	15.58	-20.2	-46.2
	KI 2nd	15.25	-19.5	-46.6	15.55	-19.8	-47.5
	KIPZ†	–	–	–	15.36	-21.0	-47.1
		E_g	W	$\langle\varepsilon_d\rangle$	E_g	W	$\langle\varepsilon_d\rangle$
GaAs	KI Full	1.74	12.7	-17.8	1.68	12.7	-16.9
	KI 2nd	1.75	12.7	-17.8	2.02	12.7	-17.4

positions of selected bands computed at full and 2nd order KI for both LiF and GaAs is given in Tab. S1. The minor differences between the KI@LDA values reported here with respect to those in the main text (compare Tab S1 with tables under Fig. 3 and Fig. 4 in the main text) are due to the different **k**-mesh (4×4×4 here vs 6×6×6 in the main text) and, for GaAs band gap, to the spin-orbit coupling correction (-0.1 eV) applied to the results in the main text (not applied here). This analysis points to a non trivial dependence of the quality of the 2nd order approximation on the system and on the base functional which requires further investigation to be deeply understood and possibly corrected toward a better agreement with the full KI method. Work in this direction is under way.

References

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